

Assignment V - PHYS 708

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Problem 1

(a) The spectral evolution of high-energy cosmic-ray electrons inside our Galaxy is governed by the following equation

$$\frac{dN(E)}{dt} = \frac{\partial}{\partial E} [b(E) N(E)] + Q(E)$$

where $N(E)$ is the cosmic-ray electron density, $b(E) = aE^2$ is the synchrotron and inverse Compton energy loss rate, and $Q(E) = \kappa E^{-p}$ is the source spectrum. For a continuous injection of electrons from sources uniformly distributed inside the whole Galactic volume, show that the spectrum inside the Galaxy at a given time t is:

$$N(E) = \begin{cases} \frac{\kappa E^{-(p+1)}}{a(p-1)} [1 - (1 - aEt)^{p-1}] & \text{if } E \leq E_0, \\ \frac{\kappa E^{-(p+1)}}{a(p-1)} & \text{if } E > E_0. \end{cases}$$

where $E_0 = 1/(at)$ is the energy at which the energy loss time becomes equal to time t .

(b) Analyse the shape of the electron spectrum at $E \leq E_0$ and $E > E_0$.

Problem. This problem involves solving the transport equation

$$\frac{\partial N(E, t)}{\partial t} = \frac{\partial}{\partial E} [aE^2 N(E, t)] + \kappa E^{-p}, \quad N(E, 0) = 0, \quad (1)$$

for a spatially uniform, continuous source switched on at $t = 0$, with $a > 0$, $\kappa > 0$, and $p > 1$ using the Laplace Transform on t . Here, the major energy losses are from synchrotron/IC leading to $b(E) = aE^2$.

Step 1: Applying the Laplace Transform in Time

We use the one-sided Laplace transform on our transport equation:

$$\bar{N}(E, s) = \int_0^\infty e^{-st} N(E, t) dt.$$

Rule 1: time derivative. With $N(E, 0) = 0$,

$$\mathcal{L}_t \left\{ \frac{\partial N}{\partial t} \right\} = s \bar{N}(E, s) - N(E, 0) = s \bar{N}(E, s).$$

Rule 2: energy derivative. E is independent of the t -transform, so

$$\mathcal{L}_t \left\{ \frac{\partial}{\partial E} [aE^2 N] \right\} = \frac{\partial}{\partial E} [aE^2 \bar{N}(E, s)].$$

Rule 3: source. $Q(E) = \kappa E^{-p}$ is time independent:

$$\mathcal{L}_t \{Q(E)\} = \int_0^\infty e^{-st} \kappa E^{-p} dt = \frac{\kappa E^{-p}}{s}.$$

Putting these into (1) gives

$$s \bar{N}(E, s) = \frac{\partial}{\partial E} [aE^2 \bar{N}(E, s)] + \frac{\kappa}{s} E^{-p}. \quad (2)$$

Next, expand the E -derivative on the right by the product rule:

$$\frac{\partial}{\partial E} [aE^2 \bar{N}] = a \left(2E \bar{N} + E^2 \frac{\partial \bar{N}}{\partial E} \right).$$

Plugging into (2) and rearranging terms to isolate $\frac{\partial \bar{N}}{\partial E}$:

$$s \bar{N} = a \left(2E \bar{N} + E^2 \frac{\partial \bar{N}}{\partial E} \right) + \frac{\kappa}{s} E^{-p} \Rightarrow aE^2 \frac{\partial \bar{N}}{\partial E} + (2aE - s) \bar{N} = -\frac{\kappa}{s} E^{-p}.$$

Finally divide by aE^2 to reach the standard linear first-order form

$$\frac{\partial \bar{N}}{\partial E} + \underbrace{\left(\frac{2}{E} - \frac{s}{aE^2} \right)}_{P(E)} \bar{N} = -\frac{\kappa}{as} E^{-(p+2)}. \quad (3)$$

Step 2: Solve the ODE in E using Integrating Factor

For $y' + P(E)y = Q(E)$, the integrating factor is

$$\mu(E) = \exp\left(\int P(E) dE\right).$$

Compute $\int P(E) dE$ term by term:

$$\int \frac{2}{E} dE = 2 \ln E, \quad \int \left(-\frac{s}{aE^2}\right) dE = -\frac{s}{a} \int E^{-2} dE = -\frac{s}{a} \left(-\frac{1}{E}\right) = \frac{s}{aE}.$$

Hence

$$\mu(E) = \exp\left(2 \ln E + \frac{s}{aE}\right) = E^2 e^{s/(aE)}. \quad (4)$$

Multiplying (3) by μ turns the left-hand side into an exact derivative:

$$\begin{aligned} \mu \bar{N}_E + \mu P \bar{N} &= \frac{d}{dE} [\mu \bar{N}] \quad (\text{by construction of } \mu), \\ \Rightarrow \frac{d}{dE} [\mu \bar{N}] &= \mu \left(-\frac{\kappa}{as} E^{-(p+2)}\right). \end{aligned} \quad (5)$$

Explicitly substituting (4) gives the RHS

$$\mu \left(-\frac{\kappa}{as} E^{-(p+2)}\right) = -\frac{\kappa}{as} E^2 e^{s/(aE)} E^{-(p+2)} = -\frac{\kappa}{as} e^{s/(aE)} E^{-p}.$$

Step 3: Integrate in E and apply the boundary condition

Integrate (5) from $E' = E$ to $E' \rightarrow \infty$:

$$\int_E^\infty \frac{d}{dE'} [\mu(E') \bar{N}(E', s)] dE' = \int_E^\infty \left(-\frac{\kappa}{as} e^{s/(aE')} E'^{-p}\right) dE'.$$

Evaluate the left as a fundamental theorem difference:

$$\lim_{E' \rightarrow \infty} \mu(E') \bar{N}(E', s) - \mu(E) \bar{N}(E, s) = -\mu(E) \bar{N}(E, s).$$

The first term vanishes because as $E' \rightarrow \infty$, the physical spectrum behaves $\bar{N} \sim E'^{-(p+1)}$ (the cooled tail), hence $\mu \bar{N} \sim E'^2 E'^{-(p+1)} = E'^{1-p} \rightarrow 0$ for $p > 1$.

Thus,

$$-\mu(E) \bar{N}(E, s) = -\frac{\kappa}{as} \int_E^\infty e^{s/(aE')} E'^{-p} dE'.$$

Divide by $-\mu(E) = -E^2 e^{s/(aE)}$ to obtain

$$\boxed{\bar{N}(E, s) = \frac{\kappa}{as} E^{-2} e^{-s/(aE)} \int_E^\infty e^{s/(aE')} E'^{-p} dE'}. \quad (6)$$

Step 4: Prepare the inverse transform (clean bounds and kernel)

Streamline the integral by $u = 1/E'$:

$$E' = \frac{1}{u}, \quad dE' = -\frac{du}{u^2}, \quad E'^{-p} = u^p.$$

Bounds: $E' : E \rightarrow \infty$ maps to $u : 1/E \rightarrow 0$. Then

$$\int_E^\infty e^{s/(aE')} E'^{-p} dE' = \int_{1/E}^0 e^{(s/a)u} u^p \left(-\frac{du}{u^2}\right) = \int_0^{1/E} e^{(s/a)u} u^{p-2} du.$$

Insert into (6):

$$\bar{N}(E, s) = \frac{\kappa}{as} E^{-2} e^{-s/(aE)} \int_0^{1/E} u^{p-2} e^{(s/a)u} du. \quad (7)$$

Now write the product of exponentials as a *time shift* with

$$\tau \equiv \frac{1}{a} \left(\frac{1}{E} - u \right) \iff u = \frac{1}{E} - a\tau, \quad du = -a d\tau.$$

As $u : 0 \rightarrow 1/E$, $\tau : 1/(aE) \rightarrow 0$. Also

$$e^{-s/(aE)} e^{(s/a)u} = e^{-s\tau}.$$

Therefore (7) becomes the clean convolution form

$$\bar{N}(E, s) = \frac{\kappa}{s} E^{-2} \int_0^{1/(aE)} \left(\frac{1}{E} - a\tau \right)^{p-2} e^{-s\tau} d\tau. \quad (8)$$

Step 5: Invert the transform and evaluate the τ -integral

Use the standard inverse pair $\mathcal{L}^{-1}\{e^{-s\tau}/s\} = H(t-\tau)$ (Heaviside function). Interchanging inversion and the τ -integral yields

$$\begin{aligned} N(E, t) &= \kappa E^{-2} \int_0^{1/(aE)} \left(\frac{1}{E} - a\tau \right)^{p-2} H(t-\tau) d\tau \\ &= \kappa E^{-2} \int_0^{\tau_{\max}(E, t)} \left(\frac{1}{E} - a\tau \right)^{p-2} d\tau, \quad \tau_{\max}(E, t) \equiv \min\left(t, \frac{1}{aE}\right). \end{aligned} \quad (9)$$

Compute the integral explicitly. Let $x = \frac{1}{E} - a\tau$ so that $dx = -a d\tau$:

$$\int \left(\frac{1}{E} - a\tau \right)^{p-2} d\tau = \int x^{p-2} \left(-\frac{dx}{a} \right) = -\frac{1}{a} \frac{x^{p-1}}{p-1} = -\frac{1}{a(p-1)} \left(\frac{1}{E} - a\tau \right)^{p-1}.$$

Apply the limits $\tau = 0$ and $\tau = \tau_{\max}$ to (9):

$$\begin{aligned} N(E, t) &= \kappa E^{-2} \left[-\frac{1}{a(p-1)} \left(\frac{1}{E} - a\tau \right)^{p-1} \right]_{\tau=0}^{\tau=\tau_{\max}} \\ &= \frac{\kappa}{a(p-1)} E^{-2} \left[\left(\frac{1}{E} \right)^{p-1} - \left(\frac{1}{E} - a\tau_{\max} \right)^{p-1} \right]. \end{aligned} \quad (10)$$

Factor $E^{-(p-1)}$ and rewrite the bracket as a power of $(1 - \dots)$:

$$\left(\frac{1}{E} \right)^{p-1} - \left(\frac{1}{E} - a\tau_{\max} \right)^{p-1} = E^{-(p-1)} \left[1 - \left(1 - aE\tau_{\max} \right)^{p-1} \right].$$

Insert into (10):

$$N(E, t) = \frac{\kappa}{a(p-1)} E^{-(p+1)} \left[1 - (1 - aE \tau_{\max})^{p-1} \right], \quad \tau_{\max} = \min\left(t, \frac{1}{aE}\right). \quad (11)$$

Step 6: Final Piecewise form

Define the cooling-break energy $E_0(t) \equiv 1/(at)$.

$$\tau_{\max} = \begin{cases} t, & E \leq E_0, \\ 1/(aE), & E > E_0, \end{cases} \implies N(E, t) = \begin{cases} \frac{\kappa}{a(p-1)} E^{-(p+1)} \left[1 - (1 - aEt)^{p-1} \right], & E \leq E_0, \\ \frac{\kappa}{a(p-1)} E^{-(p+1)}, & E > E_0. \end{cases}$$

Step 7: Shape Analysis of Electron Spectrum

Low E limit ($E \ll E_0$)

$$(1 - aEt)^{p-1} \simeq 1 - (p-1)aEt \implies N(E, t) \simeq \kappa t E^{-p} \quad (\text{uncooled injection slope}).$$

High E limit ($E \gg E_0$)

$$N(E, t) = \frac{\kappa}{a(p-1)} E^{-(p+1)} \quad (\text{cooled steady-state slope, steepened by one power}).$$

Thus the spectrum shows a cooling break at $E_0 = 1/(at)$ with indices $-p$ (below) and $-(p+1)$ (above).